

AI-Driven Grief Chatbots: Transforming Grief Management for Individuals with Chronic Illness

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Abstract

Chronic illness precipitates numerous challenges and often leading to a multifaceted grieving process. Traditional grief management approaches, such as counseling, support groups, and medication, often fall short of providing immediate and personalized support. The integration of artificial intelligence (AI) in healthcare, particularly AI-driven chatbots, offers promising enhancements for grief support. However, ethical considerations must be addressed. This study examines the application of AI-driven chatbots in managing grief among individuals with chronic illness using the Value Sensitive Design (VSD) framework and two kernel theories. By exploring and integrating human values and technical features, this research proposes guidelines for designing AI-driven grief chatbots.

Keywords: Artificial intelligence, chatbot, value sensitive design, general systems theory, dual process model of grief

1. Introduction

The loss of a healthy body precipitates numerous challenges, including altered relationships, modified career prospects, changes in hobbies, and a transformed sense of identity in the context of chronic illness (Telford et al., 2006). Grief, as the response to these challenges, constitutes a normal reaction to loss. The grieving process, especially for people with chronic illness, is multifaceted, complicated, and deeply personal (Boer et al., 2014). Chronic illness can take a psychological toll, causing symptoms such as depression, anxiety, and a deep feeling of isolation. For example, previous research has confirmed that social isolation and loneliness significantly increase the risk of depression and anxiety symptoms (Santini et al., 2020). Emotionally, those grieving may experience intense sadness, anger, or numbness. Socially, they may withdraw from friends and family, feeling that others cannot truly understand their loss (Parkes & Prigerson, 2013). This complex emotional and social response is

supported by research indicating that sadness is often accompanied by social withdrawal and a sense of isolation, which can further intensify the grieving process (Goldberg, 2008).

Traditional approaches to grief management, including counseling, support groups, and medication, have served as the foundational approaches on an individual's journey through grief (Stroebe & Schut, 1999). Counseling, both individual and group, provides an outlet for emotion and facilitates the sharing of experience and support (Neimeyer, 2016). Support groups bring together those who have experienced similar chronic illness, providing a sense of community and shared understanding (Dibb & Yardley, 2006). Pharmacotherapy can also play a role in managing symptoms of overwhelming grief by targeting related mental health issues such as depression or anxiety (Zisook & Shear, 2009).

Yet, despite their value, these interventions can fall short of addressing the unique and ever-evolving needs that accompany chronic illness. They are typically not available on-demand, and they often follow a set schedule, which may not coincide with the immediate moments of distress that are characteristic of intense grief (Holland et al., 2009). Additionally, they may offer generalized support which might not be personalized enough to accommodate the particularities of an individual's situation or needs (Di Paolo et al., 2017).

The evolution of artificial intelligence (AI) technology and its integration into healthcare field offers promising outcomes for enhancing the support available to individuals in grief. For example, CNN reported that in order to manage their bereavement and grieving process, individuals are utilizing AI to simulate the personality and behavior of deceased loved ones (Kelly, 2024). In addition, Partida (2023) proposed that AI can fix unhealthy coping mechanisms and help people work through grief. Real-time, AI-driven support systems (e.g., AI chatbot) can potentially provide several key advantages, including immediacy of response (Siala & Wang, 2022), highly personalized interventions (Al Kuwaiti et al., 2023), constantly

updates, and medium to traditional forms of care (Schueller et al., 2016).

Incorporating AI chatbot effectively into grief management also raises important considerations. The design of such systems must consider ethical implications, such as privacy concerns and the need for clear boundaries (Luxton, 2014). There is also the necessity of ensuring that AI complements rather than replaces the human element of care which is often critical in grief management (Bickmore et al., 2009).

As such, the exploration of integrating AI chatbot into grief management represents not just an addition to the toolkit of support available to those with chronic illness but a potential paradigm shift in how support is delivered and experienced. Therefore, the objective of this study is to examine the underlying human values of AI-driven chatbot in managing grief among individuals with chronic illness and provide general guidelines on designing. By applying the value sensitive design (VSD) framework and two kernel theories (General Systems Theory and Dual Process Model), this research identified values in the process of grief management and explored their conceptualizations and contextualization in chronic illness. In addition, a case study of four example AI-driven chatbots illustrated the mapped values and grief coping stressors. By interacting with human values and technical features, this research provided future empirical suggestions for designers to enhance the implications of substantial human values in AI chatbot and grief management process.

This study broadens the application of VSD to encompass the distinct needs of individuals grappling with chronic illnesses. The nature of chronic conditions often leads to complex emotional and psychological challenges, such as grief, which require careful consideration in the design and deployment of supportive technologies. This research sets a substantive precedent, elaborating on the multi-dimensional aspects of VSD engagement within this domain. Also, this research contributes guidelines for designing AI-driven chatbots tailored to the grieving processes of those affected by chronic illness. Such guidelines are proposed to help care providers and designers to create AI-driven grief chatbot.

2. Literature Review

2.1. AI-driven Chatbots in Healthcare

The integration of AI in healthcare has experienced significant growth in recent years, driven by the need to enhance patient care and streamline medical processes (Al Kuwaiti et al., 2023). Among the various applications of AI, the use of AI-driven chatbots to support patients with chronic illnesses has received

considerable attention (Haque et al., 2023). These chatbots, equipped with natural language processing (NLP) and machine learning capabilities, offer personalized, real-time support to patients, assisting with symptom management, medication adherence, and emotional well-being (Islam et al., 2023).

A growing body of research underscores the utility of AI-driven chatbots in providing ongoing support and management for chronic illnesses. For instance, Miner et al. (2019) examined chatbots' efficacy in administering cognitive behavior therapy to patients with chronic mental health issues, resulting in notable self-reported improvements. In addition, Bickmore et al. (2018) highlighted the role of conversational agents in instigating positive behavior modifications in diabetes and heart disease management. These chatbots can provide conversational interfaces that respond to user inputs with a human-like understanding, offering personalized healthcare guidance, reminders for medication intake, and an empathetic virtual presence for emotional support.

Meanwhile, there are bunch of literature review studies on AI chatbots promoting health behavioral changes. For example, Kurniawan et al. (2024) emphasizes the versatility of chatbots in preventing, treating, and aiding in the rehabilitation of chronic conditions by means of enabling self-management and providing continuous support. Also, Aggarwal et al. (2023) suggested that chatbots have been instrumental in chronic condition management through applications that extend beyond medication adherence to include weight management and lifestyle advice. The nuances of chatbot platform deployment provide a profound understanding of their capabilities and adaptation to user behaviors. Furthermore, Haque et al. (2023) examined the transformation of chronic disease management utilizing chatbots. This research provided key use cases for personalized and cost-effective care, addressing symptom monitoring, medication management, and lifestyle modifications that align with the rising demand for personalized healthcare solutions. Additionally, Bin Sawad et al. (2022) emphasized the promising future of health care chatbots in aiding the self-management of chronic conditions but noted the paucity of evidence on the long-term efficacy of these AI-powered systems.

2.2. AI-driven Chatbot in Grief Support

There are not many studies that investigate the facilitation of AI-driven chatbots in grief support. For instance, Stanciu (2023)'s thesis explored the complex interplay between AI chatbots and human emotions, notably in the context of grief. The research hypothesized that an awareness of the chatbot's presence could significantly affect the user's

engagement level, particularly during episodes of sadness or happiness. This investigation reveals insights into how AI chatbots might be programmed to respond to emotional cues and offer support to individuals experiencing grief due to chronic illness. Additionally, Xygykou et al. (2023) found that participants using simulation and companion chatbots experienced varying levels of support during their grief. The chatbot helped them reprocess feelings and explore self-identity, complementing their real-life social interactions rather than replacing them. As suggested by Vaidyam et al. (2019), chatbots have the potential to cater to individuals experiencing distress and isolation and help manage emotions, including the grief associated with chronic illnesses, thus offering an avenue for emotional support and counseling.

2.3. Technical Features of AI-driven Chatbots

AI-driven chatbots in healthcare represent a transformative intersection of technology and medicine, with a broad range of technical features and capabilities that are constantly evolving. Key technical features of AI-driven chatbots in healthcare include symptom tracking, personalized health education, medication adherence promotion, and psychological support (Islam et al., 2023; Martins et al., 2024).

In the chronic illness context, symptom checkers like Buoy Health use AI algorithms to guide patients through a clinical-grade interview process, approximating the preliminary triage typically performed by a general practitioner. These virtual interviews allow for earlier detection of potential health issues, enabling timely medical intervention (Carmona et al., 2022). Advanced chatbots also integrate with electronic health records (EHRs) to tailor interactions based on patients' medical histories, enhancing both relevance and accuracy of the information provided (Balcombe, 2023).

Psychologically, chatbots like Tess and Woebot deliver conversational therapy using cognitive behavioral therapy (CBT) techniques, crucial for managing the mental health aspects of chronic illnesses (Fulmer et al., 2018). Advanced predictive analytics enable chatbots to anticipate potential health issues, suggesting preventive actions based on user health data patterns (Podrazhansky et al., 2020).

Besides, machine learning allows these chatbots to continuously improve, becoming more responsive and accurate by recognizing individual user preferences and needs, thereby enhancing user engagement and satisfaction (Rajesh et al., 2023). Multi-language support ensures accessibility for a diverse global patient population, providing equitable healthcare support (Tresner-Kirsch et al., 2023).

Moreover, sentiment analysis capabilities enable chatbots to understand and respond empathetically to users' emotional states, offering appropriate support during times of distress, fear, or other significant emotions related to chronic illness (Mehta et al., 2022). These features collectively enhance the effectiveness and reach of AI chatbots in healthcare.

As AI chatbots become more integrated into the healthcare process, it is crucial to assess not just their functional capabilities but also their impact on the human values. Only through such a thoughtful approach can AI-driven chatbots truly enhance chronic illness grief management, merging the strengths of technology with the irreplaceable qualities of human care. To balance both technical and human values, we adopt a VSD approach to better understand AI-driven chatbot in chronic illness grieving process.

3. Theory

3.1. Value Sensitive Design

The VSD framework acts as a foundational theoretical guide for this investigation. As posited by Friedman et al. (2013), VSD proves to be a significant tool for examining the infusion of human values into information systems from both a conceptual and a technological vantage point, as well as an empirical one. At the center of this interaction between humans and information systems values—integral and universally significant to user experience. Among the many values highlighted, thirteen dominate due to their universal relevance in information system utilization: accountability, autonomy, calmness, courtesy, environmental sustainability, freedom from bias, human welfare, identity, informed consent, ownership, property, privacy, trust, and universal usability (Friedman et al., 2013).

Expanding into the healthcare domain, and specifically the chronic illness context, researchers such as Dadgar & Joshi (2018) employed the VSD framework to conduct an in-depth field study focused on the role of information and communication technology (ICT) in chronic disease self-management. Their empirical investigation is instrumental for delineating the values that are critical from the standpoint of patients grappling with enduring health conditions, highlighting the necessity for a user-centered approach in technological development. Additionally, Maathuis et al. (2020) presented a design approach that incorporates human values in the development of a web-based Quality of Life (QoL) instrument for individuals with mental health problems. They found that mental health, often a chronic and ongoing issue, requires sensitive consideration of

patient values to ensure technology designs resonate with users and support their overall well-being. Besides, Cruz-Martínez et al. (2021) applied VSD towards creating eHealth technologies to support the self-management of cardiovascular diseases, a critical area in chronic illness where user values can greatly influence the design and adoption of supportive technologies.

The VSD framework suggests that technology by its very nature is not neutral; rather, it reflects and reinforces particular human values, oftentimes those of the designers or prevailing cultural norms. In chronic disease management, where patients' daily lives are intricately linked with their information systems, the alignment of these systems with their values is crucial. By embracing VSD, this study does not only elucidate what values are deemed vital but also aims to ensure these values are carefully embedded in the design and implementation of AI-driven chatbots for chronic illness grief process self-management.

3.2. General Systems Theory - Informed Conception of an IS Artifact

Technology, to be fully understood and effectively integrated, must be viewed not as an isolated tool but as one roots in a social system. This perspective suggests that the essence of technology cannot be grasped in isolation from the societal and human dimensions that it interacts with and influences (Orlikowski & Iacono, 2001; Winter et al., 2014). Within the IS discipline, the term 'IS artifact' is a construct used to describe the tangible embodiment of an IS within an organization. Despite its widespread usage, the exact parameters defining an IS artifact remain somewhat nebulous. Addressing this conceptual ambiguity, Chatterjee et al. (2017) turned to general system theory (GST) to refine the notion of what constitutes an IS artifact. They proposed a conceptual model that situates the IS artifact at the confluence of three intertwined subsystems: the technical subsystem, which encompasses the tools and processes; the social subsystem, which refers to the human actors and organizational context; and the information subsystem, which involves the data and content transmitted by the technology. These subsystems are not static but engage in a constant interplay, striving towards a state of dynamic equilibrium (Chatterjee et al., 2017, 2021; Galliers, 2003).

Accordingly, AI-driven chatbot is not merely a conversational agent but an IS artifact itself, embedded with technical, social, and informational aspects that mirror the framework outlined by Chatterjee et al. (2017). Building upon their GST-informed model, this study articulated an expanded conceptualization of AI-

driven chatbot. This novel approach recognizes AI-driven chatbot as a complex entity that must be managed within the interconnected system of an organization's technology, personnel, and the information it governs. Our extended version provides a richer, systems-oriented view of AI-driven chatbot, offering a holistic understanding that aligns with the multifaceted nature of modern information system environments.

Specifically, there are three subsystems in the role of AI-driven chatbot. These subsystems are significant components that not only serve as foundational cornerstones but also function as facilitators for the efficacy of IS (Camarinha-Matos & Afsarmanesh, 2003). The first is the social subsystem, which encapsulates the users (chronic illness patients) who interact with the AI-driven chatbots. This subsystem embodies the human element, comprising the individuals and the societal frameworks within which they engage with chatbots. Its essence is characterized by the complex web of social dynamics, behaviors, and expectations that shape the user experience.

Next, the technological subsystem represented by the AI-driven conversational agents themselves. This subsystem includes the specific technologies, such as artificial intelligence and machine learning tools, along with the facilitating infrastructure like applications, computational hardware, and network connectivity that are integral to the operation of these IS artifacts.

Lastly, the information subsystem encompasses the data-centric components—namely the grief-related information and resources that are communicated and processed by the chatbot. This subsystem is the repository for the relevant knowledge, from the nuances of individual grief experiences to broader information on illness and bereavement, as well as personal data.

The interaction among these subsystems is neither isolated nor static but is characterized by mutual reinforcement and continuous entwinement (Chatterjee et al., 2021). When users express their needs related to grief, these needs serve as inputs which initiate a collaborative response from all three subsystems. The emergent outputs—the tangible results of this interaction—aim to enhance well-being, reflecting the overarching goal of the chatbot's function. As Chatterjee et al. (2017) articulately stated, all subsystems (social, technical, and information) are adaptable, delineated by boundaries that are neither rigid nor impermeable, but rather evolve to align with shifts in contextual circumstances and mechanisms (Chatterjee et al., 2017). Feedback loops, stemming from user and practitioner interactions, fuel the continuous evolution of inputs and outputs, prompting the dynamic operation of the three subsystems to accommodate and synchronize these changes.

Moreover, given the individual differences among users, there is a spectrum of unique inputs and resultant outputs that feed back into the system, influencing the interactivity and transformation of the subsystems. For instance, a user's personal coping style or support network may have distinct implications for their use of the chatbot, leading to customized utilization patterns and outcomes. The feedback and results garnered from each user further refine the AI's approach, ensuring a sensitive, tailored experience for each individual.

This conceptual framework not only explicates how an IS artifact functions in general terms but also illuminates the specific process by which AI-driven chatbots deliver bespoke support to individuals coping with grief, dynamically adapting to the needs and contexts of chronic illness patients.

3.3. Dual Process Model of Grief

The Dual Process Model (DPM) of grief, developed by Stroebe & Schut (1999), posits that effective coping with bereavement involves oscillating between confronting and avoiding grief-related stressors, known as loss-oriented and restoration-oriented activities, respectively, each of which plays a pivotal role in shaping a person's experience of loss and adaptation.

Loss-oriented stressors refers to the essence of the loss experienced. In the context of chronic illness, these stressors manifest as the deep sense of loss tied to the altered state of one's body, coupled with the symptoms that accompany it (Charmaz, 2006). This loss is profound; it encompasses coming to terms with the significant changes in bodily function, physical appearance, and vitality (Hewson, 1997). Essentially, it's a mourning for the life that could have been (without illness) and the acceptance of a new reality contoured by the limitations that the illness introduces (Dickson et al., 2008).

In contrast, restoration-oriented stressors represent the secondary challenges from the primary loss. This category of stressors interrogates the modifications one must make to their lifestyle to adapt their illness (Helm et al., 2007). It looks at how the illness will change one's vocational capacity, recreational activities, social dynamics, and the adaptability required in everyday life routines (Zautra, 1996). These stressors necessitate a constructive reassembly of one's life, accounting for the disarray caused by illness and cultivating a new form of identity (De Ridder et al., 1998).

Engaging with loss-oriented stressors typically about embracing and enduring the intense emotions that stem from the permanence of the loss. It involves a deep yearning for a return to a pre-loss state and wrestling with feelings of anger and injustice due to the unwelcome changes imposed by chronic illness. The

process of coping becomes an exercise in vulnerability—allowing oneself to fully experience and process the profound feelings of grief (Eisma et al., 2022).

Conversely, addressing restoration-oriented stressors implies the proactive reconstruction of a post-loss existence, by changing career trajectories, remolding roles within relationships, and reinventing daily life practices. Effective coping is characterized by the commitment to looking forward, developing new trajectories that weave the presence of illness (Helm et al., 2007).

Stoebe and Schut (1999) assert that a balanced grieving process demands acknowledgement and engagement with both types of stressors. An overemphasis on loss-oriented grief risks trapping one in a past that cannot be revived, while an exclusive focus on restoration-oriented grief risks severing ties with the past, hindering the integration of that past into a new identity. The essence of 'good grief,' then, is found in the attentive navigation between these stressors.

Oscillation captures the dynamic essence of the grieving process; it's a motion between addressing the immediate sting of loss and engaging in the reconstruction of life in the wake of it (Hooghe et al., 2012). This oscillation is not a linear pathway but a complex journey, fluid and deeply personal. At one point, an individual might find themselves confronting painful reminders of their loss—perhaps a reflection of physical changes in the mirror—only to pivot in the next breath towards proactive strategies, such as selecting attire that celebrates their body's new narrative (Monteiro, 2014).

At the core of the DPM is the acknowledgment that grief is not a static, one-size-fits-all experience—it's a multifaceted journey interspersed with periods of immersion in loss, adaptation, and repose. The DPM contextualizes the full lens of grieving experiences, illustrating that people can be at once deeply affected by their loss, actively engaged in the recalibration of their lives, and also maintain aspects of their identity that remain distinct from grief.

4. Method

4.1. Data Collection

We conducted a systematic examination of several scholarly databases, including Business Source Complete (EBSCO), AIS Library, and Google Scholar. The scope of our research was defined using a series of targeted search terms: chatbot, AI-driven chatbot, conversational agent, and AI. To enhance the rigor and validity of our study, we carefully selected 85 scholarly

papers that were most relevant for our review and thematic analysis.

4.2. Identified Values

Building upon the work of Friedman et al. (2008), this research employed an array of 13 values as a baseline to categorize and synthesize the various value dimensions revealed in the review. These values served as a lens through which we could discern the underpinning ethical implications within the literature concerning AI-driven grief chatbot.

Drawing insights from the GST and DPM, this study distilled the findings further to identify seven core values that demonstrate a direct relationship with AI-driven grief chatbots in the context of chronic illness and IS field. These central values were examined in detail, and their corresponding definitions were provided in Table 1.

Values	Original Definition	Contextualized Definition
Autonomy	Refers to people's ability to decide, plan, and act in ways that they believe will help them to achieve their goals (p.58)	User's capacity to engage with the chatbot in a way that they feel supports their unique grieving process.
Calmness	Refers to a peaceful and composed psychological state (p.59)	The interactions are designed to soothe the user, minimize anxiety or agitation associated with grief, and provide a virtual space for reflection and peace.
Informed Consent	Refers to garnering people's agreement, encompassing criteria of disclosure and comprehension (for "informed") and voluntariness, competence, and agreement (for "consent") (p.58)	Ensuring that users understand the capabilities of the AI-driven grief chatbot, the extent to which their data and interactions are used, and the purpose behind them.
Privacy	Refers to a claim, an entitlement, or a right of an individual to determine what information about himself or herself can be communicated to others (p.58)	How their personal and sensitive information is collected, used, and shared by the chatbot providers.
Courtesy	Refers to treating people with politeness and consideration (p.58)	Chatbot's programming to interact with users with kindness, respect, and sensitivity.
Human Welfare	Refers to people's physical, material, and psychological well-being (p.58)	Emotional support that may alleviate psychological distress and provide comfort.
Trust	Refers to expectations that exist between people who can experience good will, extend good will toward others, feel vulnerable, and experience betrayal (p.58)	User's expectation that the chatbot will function in a benevolent manner.

Table 1. The definitions of each value

5. Case Analysis

AI-driven chatbots specifically designed for grief management in chronic illness are currently rare. However, several existing types of healthcare-related chatbots have features that could be beneficial for this purpose. We have identified four categories of AI-driven chatbots that, while not explicitly created for grief management in chronic illness, offer functionalities that could support individuals in this situation.

This study established the following criteria for including chatbots in our analysis: a) Relevance: The chatbot must have features applicable to grief management or chronic illness support. b) AI-driven: The system must utilize AI technologies. c) Accessibility: The chatbot must be publicly available or have sufficient public documentation for analysis. d) Diversity: The selection should represent a range of approaches and features.

5.1. Types

5.1.1. AI-Enhanced Bereavement Support Chatbots.

Chatbots are designed to offer bereavement counseling, emotional support, and stress management strategies that can be particularly helpful for individuals dealing with grief on top of managing a chronic illness. For example, Grief Works (<https://illumapps.com/griefworks>) sends personalized messages to people who are grieving, with advice and support that's tailored to their loss and timing.

5.1.2. AI-Driven Mental Health Companions. While not necessarily grief-specific, these chatbots offer mental health support and could be adapted to provide assistance to those who are grieving. Companies like Woebot Health (<https://woebothealth.com>) have developed AI-driven chatbots that serve as a "guided self-help ally." It uses cognitive-behavioral therapy principles to engage with users and help them with their mental health, including grief-related issues.

5.1.3. Preventive Care Chatbots. These chatbots are engineered to facilitate proactive health management while simultaneously attending to the psychosocial needs of users, inclusive of those experiencing grief in the context of chronic illness. Innovative AI chatbot solutions, exemplified by Eleos Health (<https://eleos.health>), elevate preventive care by integrating health coaching and psychological support modules.

5.1.4. AI-Powered Therapeutic Chatbots. These are designed to simulate conversation and provide therapeutic responses, which can be useful for those who need to talk about their grief and feelings regarding their chronic illness. For instance, Tess by X2 AI

(<https://www.x2ai.com/individuals>) is a psychological AI that administers highly personalized psychotherapy, psychoeducation, and health-related reminders based on the emotional content of the conversation.

5.2. Features and Values Mapping

Based on the mentioned exemplary chatbots, the study examined their technical attributes alongside the human values. The previous discussion suggested seven key human values that are intrinsic to AI-driven chatbots: autonomy, informed consent, privacy, calmness, courtesy, human welfare, and trust. Accordingly, we have executed a thorough mapping of these chatbots' functionalities to the corresponding values they encompass (Table 2).

Feature of Chatbot	Grief Works	Woebot Health	Eleos Health	Tess by X2 AI
Autonomy	User-controlled support	User-driven conversation	Provider workflow choice	User-initiated interaction
Calmness	Soothing experience	N/A	N/A	Supportive conversations
Informed Consent	Clear service information	N/A	N/A	N/A
Privacy	Confidential experience	Secure platform	Encrypted documentation	Secure communications
Courtesy	Caring service tone	N/A	Provider burnout relief	Friendly interaction
Human Welfare	N/A	Mental well-being focus	Quality of care focus	Preventive care aim
Trust	N/A	Evidence-based approach	Clinically vetted AI	N/A

Table 2. Features and values mapping

Meanwhile, with the guidance of DPM, this research mapped the practical examples to the two distinct stressors associated with grieving: loss-oriented stressors and restoration-oriented stressors (Table 3).

Chatbot Example	Loss-Oriented Stressors	Restoration-Oriented Stressors
Grief Works	Emotional processing of grief (fear, anger, sadness, guilt, loneliness)	Tools for rebuilding life and adapting to change
Woebot Health	Emotional well-being addressing through CBT (indirect relation to grief)	Cognitive-behavioral techniques for managing feelings and promoting coping
Eleos Health	Indirect through provider support (more time for patient care)	Automation for practitioners to support care adaptation and patient coping
Tess by X2 AI	Emotional support and processing for a variety of feelings	Coping mechanisms for adapting and managing new realities

Table 3. Features and stressors mapping

This analysis yielded several key insights. These insights formed the foundation for developing our design guidelines, ensuring that they are grounded in the observed patterns and gaps in existing chatbot implementations.

Value Integration: While all chatbots incorporated multiple values, none comprehensively addressed all seven identified values. This suggests an opportunity for

more holistic chatbot design that consciously integrates all relevant human values.

Stressor Balance: The analysis revealed that most chatbots tend to focus more on restoration-oriented stressors than loss-oriented stressors. This imbalance indicates a potential area for improvement in chatbot design, as addressing both types of stressors is crucial for effective grief management.

Feature-Value Alignment: Certain features consistently aligned with specific values. For instance, user-controlled interactions were strongly associated with autonomy, while secure platforms consistently supported privacy. This insight helps establish clear connections between technical features and human values.

Contextual Adaptability: The analysis highlighted the importance of contextual adaptability in chatbot design. Features that allow for personalization and user-driven interactions were particularly effective in addressing the unique needs of individuals with chronic illnesses.

6. Guidelines

Autonomy-Respecting Adaptations: In developing AI-driven grief chatbots, it is crucial to build systems that honor user autonomy by providing adaptive response mechanisms. The chatbot should offer users the ability to guide conversations according to their personal grief journey, enabling them to decide, plan, and act on the support they need, in alignment with their goals and emotional states.

Calmness-Inducing Engagement: Our analysis showed that while some chatbots incorporated calmness-inducing features, this value was not consistently addressed across all platforms. Given the emotional volatility often associated with grief, particularly in the context of chronic illness, we identified this as a crucial area for improvement. The chatbot's proactive check-ins must be designed to embody calmness—a peaceful and composed psychological state. Interactions should be soothing and reassuring, contributing positively to the user's emotional climate.

Informed Consent and Privacy: The examination of existing chatbots highlighted varying degrees of attention to informed consent and privacy. Given the sensitive nature of grief and health information, we determined that clear explanations about data use and robust privacy measures should be standard features in grief chatbot design. The design and operation of the chatbot must prioritize informed consent and privacy. Users should be fully briefed on how their data is used, with clear explanations provided about the chatbot's functions. Ensuring privacy involves securing all user

interactions and personal information, upholding their right to determine what information about themselves is communicated and stored.

Courtesy-Centric Communication: The analysis revealed that chatbots with caring service tones and friendly interactions were better received by users. This insight led to the recommendation for consistently respectful and empathetic language in all chatbot communications. Every interaction with the grief chatbot should be imbued with courtesy, treating users with utmost politeness and consideration. The language and tone adopted by the chatbot must be consistently respectful and empathetic, especially considering the sensitive nature of the users' experiences.

Resource Provision with Human Welfare in Mind: Our mapping showed that chatbots focusing on mental well-being and quality of care had a positive impact on human welfare. This observation led to the guideline emphasizing the provision of diverse resources to support users' overall well-being. The grief chatbot should provide a range of resources with the user's physical, material, and psychological well-being as the central focus. This includes an informative content library about grief, access to local and global support networks, and direct referrals to professional services—ensuring a holistic approach to addressing the varied aspects of the user's welfare.

Trust-Embedded Scalability and Cultural Sensitivity: The analysis highlighted that evidence-based approaches and clinical vetting were key trust-building mechanisms. Extending this insight, we recommend incorporating cultural sensitivity to build trust across diverse user groups. As the chatbot aims to scale and embrace cultural diversity, a foundation of trust is paramount. Developing multilingual capabilities and cultural sensitivities into the design ensures that the chatbot respects different mourning practices and expressions of loss. By doing so, the chatbot caters to a more extensive demographic effectively and respectfully, establishing itself as a trustworthy source of solace and support.

Learning and Evolution with Human Welfare and Trust in Focus: A commitment to continuous learning, grounded in human welfare and trust, enhances the chatbot's effectiveness. Feedback loops should not only refine the chatbot's functionality but also reassure users that the system is responsive to their needs and evolving with their best interests in mind. This consistent refinement is essential in maintaining the chatbot as a reliable, empathetic, and evolving source of grief support.

7. Conclusion

This research advances the understanding of AI-driven grief chatbots within the context of chronic illness and enriches the field of IS healthcare design by employing a rigorous design science methodology (VSD). Grounded in a thorough literature review, the study identifies seven fundamental human values, namely autonomy, informed consent, privacy, calmness, courtesy, human welfare, and trust. Through a detailed examination of four exemplary AI-driven grief chatbots in practice, the investigation highlights the human values each chatbot captures and the stressors they aim to alleviate. Drawing on these findings, the research provides general design guidelines for the development of grief chatbots, each guideline directly linked to the insights gained from the analysis. This study opens avenues for future research in AI-driven grief support, emphasizing longitudinal efficacy, cross-cultural adaptations, and ethical considerations. The findings provide a framework for developers and healthcare providers, inform policymaking, and offer insights for customizing AI chatbots to address grief in chronic illness contexts.

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